

CS23334 Fundamentals of Data Science Lab

Experiment 11

Data Frame Use Cases & Operations in Data Science using Pandas

Objective: To understand and implement key DataFrame creation, column assignment, CSV dataset loading (50_Startups and Employee datasets), Series statistical operations, and array-DataFrame conversions in Pandas.

Methodology:

- 1 **DataFrame Construction & Column Assignment:** Construct DataFrames from Python lists and explicitly define column headers (EmpId, Name, Salary).
- 2 **Dataset Loading & Structural Inspection:** Import CSV datasets (50_Startups.csv, employee.csv) and inspect structure using `.info()`, `.head()`, and `.tail()`.
- 3 **Statistical & Summary Analysis:** Extract Series objects and compute summary statistics including `mean()`, `median()`, `mode()`, `var()`, `std()`, and multi-attribute `.describe()`.
- 4 **Array Conversion & DataFrame Reconstruction:** Extract column names (`.columns`) and underlying NumPy array values (`.values`), and reconstruct DataFrames.

Code:

```
import numpy as np
import pandas as pd

list_data = [[1, 'Smith', 50000], [2, 'Jones', 60000]]

df = pd.DataFrame(list_data)
df

df.columns = ['Empd', 'Name', 'Salary']
df

df.info()

df_startups = pd.read_csv("50_Startups.csv")
df_startups.info()
df_startups.head()
df_startups.tail()

df_emp = pd.read_csv("employee.csv")
df_emp.head()
df_emp.tail()
df_emp.info()

df_emp.salary
type(df_emp.salary)

df_emp.salary.mean()
df_emp.salary.median()
df_emp.salary.mode()
df_emp.salary.var()
df_emp.salary.std()

df_emp.describe()
df_emp.describe(include='all')

empCol = df_emp.columns
emparray = df_emp.values

employee_DF = pd.DataFrame(emparray, columns=empCol)
employee_DF
```

OUTPUT: 1. Manual DataFrame & Dataset Structural Inspection

```
In [1]: df = pd.DataFrame([[1, 'Smith', 50000], [2, 'Jones', 60000]])
```

```
In [2]: df.columns = ['Empd', 'Name', 'Salary']
```

```
In [3]: df
```

```
Out[3]:
```

	Empd	Name	Salary
0	1	Smith	50000
1	2	Jones	60000

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 2 entries, 0 to 1
```

```
Data columns (total 3 columns):
```

#	Column	Non-Null Count	Dtype
0	Empd	2 non-null	int64
1	Name	2 non-null	object
2	Salary	2 non-null	int64

```
dtypes: int64(2), object(1)
```

```
memory usage: 176.0+ bytes
```

```
In [5]: df_startups = pd.read_csv("50_Startups.csv")
```

```
In [6]: df_startups.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 50 entries, 0 to 49
```

```
Data columns (total 5 columns):
```

#	Column	Non-Null Count	Dtype
0	R&D Spend	50 non-null	float64
1	Administration	50 non-null	float64
2	Marketing Spend	50 non-null	float64
3	State	50 non-null	object
4	Profit	50 non-null	float64

```
dtypes: float64(4), object(1)
```

```
memory usage: 2.1+ KB
```

```
In [7]: df_startups.head()
```

```
Out[7]:
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

OUTPUT: 2. Series Statistics & DataFrame Reconstruction Results

```
In [8]: df_emp.head()
```

```
Out[8]:
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000

```
In [9]: df_emp.salary.mean()
```

```
Out[9]: 5714.285714285715
```

```
In [10]: df_emp.salary.median()
```

```
Out[10]: 6000.0
```

```
In [11]: df_emp.salary.mode()
```

```
Out[11]:
```

```
0    5000
1    6000
```

```
Name: salary, dtype: int64
```

```
In [12]: df_emp.salary.var()
```

```
Out[12]: 2571428.5714285714
```

```
In [13]: df_emp.salary.std()
```

```
Out[13]: 1603.5674514745463
```

```
In [14]: df_emp.describe(include='all')
```

```
Out[14]:
```

	emp id	name	salary
count	7.000000	7	7.000000
unique	NaN	6	NaN
top	NaN	SRI HARSHAVARDHANAN R	NaN
freq	NaN	2	NaN
mean	4.000000	NaN	5714.285714
std	2.160247	NaN	1603.567451

```
In [15]: empCol = df_emp.columns
```

```
In [16]: emparray = df_emp.values
```

```
In [17]: employee_DF = pd.DataFrame(emparray, columns=empCol)
```

```
In [18]: employee_DF
```

```
Out[18]:
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000